DESCRIPTION OF COURSEWORK

Course Code	EEE202
Course Name	Engineers and Society
Lecturer	Ir Dr Ab Halim Bin Abu Bakar
Academic Session	09/2024
Assessment Title	Assignment 1: Engineering Practice

A. Introduction/ Situation/ Background Information

The technology development has significant impact on engineering practice

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Apply principles of ethics of a professional engineer in relation to society and norms of engineering practice.

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C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

Answer questions No 3, No 6, No 10 and No 12 topic from the list as in Appendix 11

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Understanding of topics	20
2.	Information	20
3.	Use of facts	20
4.	Analysis	20
5.	Organization	20

F. Task(s)

1) Discuss the topics aspects using evidence or literature
2) Cite and include a list of all resources or references used in your presentation

APPENDIX 1

MARKING RUBRICS

Component Title	**Engineering Practice**				**Percentage (%)**		
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Understanding of Topic	Excellent understanding of the main points of topic in depth	Good understanding of the n main point of the topic	Satisfactory understanding of the main points of the topic	Poor understanding of the main points of the topic	Not able to understand the main point of the topic	20	
Information	All information presented was clear and thorough	Most information presented was clear and thorough	Most information presented was clear but not thorough	Some information was clear but there were some minor inaccuracies	Information had some major inaccuracies or was not clear	20	
Use of Fact	Every major point was well supported with several relevant facts, statistic and/or examples	Every major was adequately supported with relevant facts statistics and/or examples	Every major was supported with facts statistics and/or examples but the relevancy of some was questionable	Some points were supported with relevant acts statistics and/or examples	All points were not supported with relevant acts statistics and/or examples	20	
Analysis	All arguments were relevant with strong evidence	Most arguments were relevant with moderate evidence	Several arguments were relevant with sufficient evidence	Most arguments were irrelevant and weak evidence	All arguments are irrelevant and no evidence	20	
Organization	All arguments were clearly related to an idea and well organized in logical order Opening statement engages the interest of audience and closing statement leaves no unanswered issues and resonates with the audience.	Most arguments were clearly related to an idea and well organized in logical order Opening statement engages the interest of audience and closing statement summarizes many arguments	Some arguments were clearly related to an idea and well organized in logical order Opening statement outlines or lists argument but does not generate interest and close statement briefly restates the ideas offered in the opening statement	Most arguments were not clearly related to an idea and organized in logical order Opening statement minimally outlines arguments and closing statement does not reflect remarks	Arguments were not related to an idea. Opening statement has no outlines arguments and closing statement does not reflect remarks	20	
					TOTAL		**100**

APPENDIX 11

Questions on Regulations on Professional Conduct

The main purpose of these questions is to provide an opportunity for the candidates to demonstrate their professionalism. A candidate should have gained some understanding of the IEM Regulations on Professional Conduct before entering for the Professional Interview.

A candidate would be expected to demonstrate: -

(a) That he has thought sufficiently about the role of the engineer in the society vis-à-vis his Professional Conduct.

(b) That he can write in clear and concise manner that is intelligible to laymen.

Candidate would be expected to answer only 1 question out of 2 selected by the Examiners from among the 13 questions listed below: -

1. As a young engineer, you are instructed by your employer to prepare a bankable report for a prospective project based on the information provided by your employer. You have no access to ascertain the reliability of the information provided. In light of the IEM Regulations on Professional Conduct, how do you advise your employer so that you may complete your assignment with integrity?

2. You are assigned by your employer to prospect an area for an extractable material. However, in the course of your work, you discover no such object material but some other valuable material which your employer is not interested. On the other hand, the information which you have is of great importance to the adjoining interest. As an engineer can you release the particular information which you have to the other party so that the other valuable material may be explored? Please discuss taking into consideration provisions of the IEM Regulations on Professional Conduct.

3. Mr. X is a project engineer for a project contracted to Y Company owned by Mr. Z. Mr. Z tells Mr. X that he is ordering the latest model of an electronic gadget for his company staff, and he can get a set for Mr. X at the company discounted price. Mr. X agrees and pays on delivery. Did Mr. X breach the IEM Regulations on Professional Conduct**:**
   "It is the responsibility of every engineer to give the highest regard to his profession and conduct himself with integrity when executing his task".

   Discuss this statement in detail within the context of the Malaysian environment.

4. Professionalism cannot be taught by regurgitating the IEM Regulations on Professional Conduct nor by memorizing a set of rules. Do you agree with the statement and why?

5. The IEM Regulations on Professional Conduct speaks against active self-promotion and advertisement. In what ways would such restraint enhance the stature of engineers as professionals? More importantly, given the advent of the information technology age where virtual reality is becoming the norm, how would any violation of the IEM Regulations on Professional Conduct be detected and dealt with?

6. You are an engineer, and it is known that your project activities/plant production had caused an adverse health, safety and/or environmental (HSE) impact. To improve the HSE quality, you must cease your project activities/plant production for a period, resulting in the inability to meet the required schedule and targets.

   How would you handle a situation where there is a conflict between the IEM Regulations on Professional Conduct and commercial consideration?

7. Very often codes of practice suggest discretion of engineering judgement. There are merits and demerits of international codes of practice and standards as opposed to national documents within the context of the Malaysian industry and practices.

   Discuss by giving examples of engineering judgement whereby such decision does not conform to an international (?) code of practice and how it is justified. Is international conformity always desirable?

8. Engineers are often entrusted with the responsibility of implementing large-scale projects involving millions of Ringgits. In view of the large amount of money changing hands, the integrity and ethical standards of engineers are being put to test. What measures can an engineer take to prevent corruption and unethical practice of cutting corners to safeguard the good image of the engineering profession?

9. 'Conflict of professional interest frequently arises in the working environment of an engineer'. Discuss the statement and propose the appropriate solutions for overcoming the problems.

10. Most projects are awarded on a competitive bidding basis. Supposing you had been the designated engineer of one of the bidders which was not awarded the letter of intent by the Client; and you are then approached by the 'winner' to join the Consultant Team; should you accept the second appointment? If yes, describe the situations, which might be seen as conflict of interests. If no, describe the reasons and/or clear-cut situations why you should decline the second appointment.

11. You are in a position of responsibility administering an institution of higher learning with limited resources. You have been assigned to revise the engineering curriculum and directed to increase the number of graduates. From the standpoint of professional conduct, describe the necessary curriculum contents which must be included so as not to compromise the engineering standards.

12. Corruption has undermined integrity in professional practice in the engineering industry. The professional engineer who steadfastly embraces the IEM Regulations on Professional Conduct is left in a dilemma. His option is that he could just ignore it or turn whistle blower. Explain how you would handle this problem.

13. Assuming that you are the principal partner of a reputable engineering-based firm in a small state in the country. Your family is very well reputed, respected and influential, having lived and worked in the state for decades.

    Leaders of a certain political party approach you are seeking your assistance to campaign for them in a forthcoming state by-election. Amongst others, they have promised upon winning the said elections, to award your company with all the major projects to be awarded by the state government. It is desirable for your firm to secure new projects. Discuss critically on your actions going forward, taking account of the IEM Regulations on Professional Conduct.